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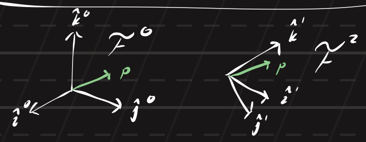
Chapter 2



$\hat{i}^i, \hat{j}^i, \hat{k}^i$ ($\cdot^i$ denotes frame)
mutually orthogonal
bases

There are different frames and we want to express a vector in one frame using the other

In $\mathcal{F}^0$ we have $\vec{p} = p_x^0 \hat{i}^0 + p_y^0 \hat{j}^0 + p_z^0 \hat{k}^0$
In $\mathcal{F}^1$ we have $\vec{p} = p_x^1 \hat{i}^1 + p_y^1 \hat{j}^1 + p_z^1 \hat{k}^1$

$$\Rightarrow \quad p_x^0 \underline{\hat{i}}^0 + p_y^0 \underline{\hat{j}}^0 + p_z^0 \underline{\hat{k}}^0 = p_x^1 \underline{\hat{i}}^1 + p_y^1 \underline{\hat{j}}^1 + p_z^1 \underline{\hat{k}}^1$$

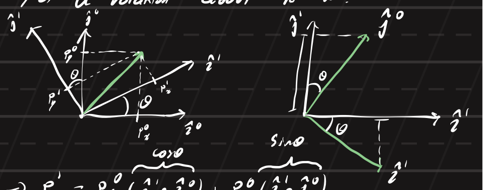
Doing a dot prod on each side with the bases of $\mathcal{F}^1$:

$$p_x^1 \underbrace{\hat{i}^1 \cdot \hat{i}^1}_{1} + p_y^1 \underbrace{\hat{j}^1 \cdot \hat{i}^1}_{0} + \ldots = (p_x^0 \hat{i}^0 + \ldots)\, \hat{i}^1$$

going from frame 0 to 1

$$\Rightarrow \vec{p}^{\,1} = \begin{bmatrix} p_x^1 \\ p_y^1 \\ p_z^1 \end{bmatrix} = \begin{bmatrix} \hat{i}^1 \cdot \hat{i}^0 & \hat{i}^1 \cdot \hat{j}^0 & \hat{i}^1 \cdot \hat{k}^0 \\ \hat{j}^1 \cdot \hat{i}^0 & \hat{j}^1 \cdot \hat{j}^0 & \hat{j}^1 \cdot \hat{k}^0 \\ \hat{k}^1 \cdot \hat{i}^0 & \hat{k}^1 \cdot \hat{j}^0 & \hat{k}^1 \cdot \hat{k}^0 \end{bmatrix} \begin{bmatrix} p_x^0 \\ p_y^0 \\ p_z^0 \end{bmatrix} \Rightarrow \vec{p}^{\,1} = R_0^1 \vec{p}^{\,0}$$

For a rotation about $\hat{k}^0 = \hat{k}^1$:



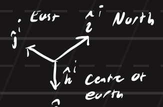
Basically projecting the original basis onto the new basis gives the factors

$$\Rightarrow p_x^1 = p_x^0 \overbrace{(\hat{i}^1 \cdot \hat{i}^0)}^{\cos\theta} + p_y^0 \overbrace{(\hat{i}^1 \cdot \hat{j}^0)}^{\sin\theta}$$

$$p_y^1 = p_x^0 \underbrace{(\hat{j}^1 \cdot \hat{i}^0)}_{-\sin\theta} + p_y^0 \underbrace{(\hat{j}^1 \cdot \hat{j}^0)}_{\cos\theta}$$

Frames:

$\mathcal{F}^i$: inertial, earth axes

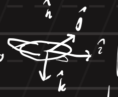


NED

$\mathcal{F}^v$: Vehicle, adjusts for pos.

aligned w/ NED

$\mathcal{F}^{v1}$: vehicle frame 1, adjusts for heading



rotation about $\hat{k}^v$ by heading $\Rightarrow \vec{p}^{\,v1} = R_v^{v1}(\psi)\, \vec{p}^{\,v} = \begin{bmatrix} \cos\psi & \sin\psi & 0 \\ -\sin\psi & \cos\psi & 0 \\ 0 & 0 & 1 \end{bmatrix} \vec{p}^{\,v}$

$\mathcal{F}^{v2}$: vehicle frame 2, adjust for pitch ...

rotation about $\hat{j}^{v2} \Rightarrow p^{v2} = R_{v1}^{v2}(\theta)\, \vec{p}^{\,v1} = \begin{bmatrix} \cos\theta & 0 & -\sin\theta \\ 0 & 1 & 0 \\ \sin\theta & 0 & \cos\theta \end{bmatrix} \vec{p}^{\,v1}$

$\mathcal{F}^b$: body, adjust for roll ...

rotation about $\hat{i}^{v2} \Rightarrow p^b = R_{v2}^b(\phi)\, \vec{p}^{\,v2} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\phi & \sin\phi \\ 0 & -\sin\phi & \cos\phi \end{bmatrix} \vec{p}^{\,v2}$

$$\Rightarrow \vec{p}^{\,b} = R_{v2}^b(\phi)\, R_{v1}^{v2}(\theta)\, R_v^{v1}(\psi)\, \vec{p}^{\,v}$$

$\mathcal{F}^s$: Stability, $\hat{i}$ is "aligned" with flow, for angle of attack

$\mathcal{F}^w$: wind, for sideslip

Velocity:

V_g – inertial velocity (ground view)
V_a – Vel relative to air
V_w – wind velocity

$$\vec{V}_a = \vec{V}_g - \vec{V}_w$$

$$\Rightarrow V_g^b = \begin{bmatrix} u \\ v \\ w \end{bmatrix} \Big\} \text{ inertial velocity in body frame}$$

$$V_w^b = \begin{bmatrix} u_w \\ v_w \\ w_w \end{bmatrix} \Big\} \text{ wind velocity in body frame} = R_v^b(\phi, \theta, \psi) \begin{pmatrix} w_n \\ w_e \\ w_d \end{pmatrix}$$

$$V_a^b = \begin{bmatrix} u_r \\ v_r \\ w_r \end{bmatrix} = \begin{bmatrix} u - u_w \\ v - v_w \\ w - w_w \end{bmatrix} \Big\} \text{ relative airspeed in body frame}$$

From a similar analysis to above, using the definition of V_a and rotating to the body frame:

$$\begin{bmatrix} u_r \\ v_r \\ w_r \end{bmatrix} = V_a \begin{bmatrix} \cos\alpha\cos\beta \\ \sin\beta \\ \sin\alpha\cos\beta \end{bmatrix}$$

Very important for physics

$$\Rightarrow V_A = \sqrt{u_r^2 + v_r^2 + w_r^2}$$

$$\alpha = \tan^{-1}\left(\frac{w_r}{u_r}\right)$$

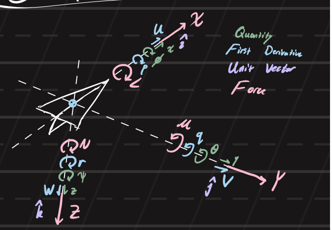
$$\beta = \sin^{-1}\left(\frac{v_r}{\sqrt{u_r^2 + v_r^2 + w_r^2}}\right)$$

Derivatives

$$\frac{d}{dt_i}\vec{p} = \frac{d}{dt_b}\vec{p} + \vec{\omega}_{b/i} \times \vec{p} \Big\} \text{ angular vel of body frame relative to inertial}$$

1.) Set up simple rendering,
2.) Implement rotation functions ⇒ one function taking angles and outputting rotation matrix
3.) Test translation and rotation

Chapter 3



Kinematic relations

$$\frac{d}{dt}\begin{bmatrix} p_n \\ p_e \\ p_d \end{bmatrix} = R_b^v \begin{bmatrix} u \\ v \\ w \end{bmatrix} = \left(R_v^b\right)^T \begin{bmatrix} u \\ v \\ w \end{bmatrix}$$

each angle is in a diff FOR so we diff in that FOR and then rotate

$$\begin{bmatrix} p \\ q \\ r \end{bmatrix} = \begin{bmatrix} \dot\phi \\ 0 \\ 0 \end{bmatrix} + R_{v_2}^b(\phi)\begin{bmatrix} 0 \\ \dot\theta \\ 0 \end{bmatrix} + R_{v_2}^b(\phi)\, R_{v_1}^{v_2}(\theta)\begin{bmatrix} 0 \\ 0 \\ \dot\psi \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & -\sin\theta \\ 0 & \cos\phi & \sin\phi\cos\theta \\ 0 & -\sin\phi & \cos\phi\cos\theta \end{bmatrix}\begin{bmatrix} \dot\phi \\ \dot\theta \\ \dot\psi \end{bmatrix}$$

$$\begin{bmatrix} p_n \\ p_e \\ p_d \end{bmatrix} = \begin{bmatrix} \cos\theta\cos\psi & \sin\phi\sin\theta\cos\psi - \cos\phi\sin\psi & \cos\phi\sin\theta\cos\psi + \sin\phi\sin\psi \\ \cos\theta\sin\psi & \sin\phi\sin\theta\sin\psi + \cos\phi\cos\psi & \cos\phi\sin\theta\sin\psi - \sin\phi\cos\psi \\ -\sin\theta & \sin\phi\cos\theta & \cos\phi\cos\theta \end{bmatrix}\begin{bmatrix} u \\ v \\ w \end{bmatrix}$$

$$\begin{bmatrix} \dot\phi \\ \dot\theta \\ \dot\psi \end{bmatrix} = \begin{bmatrix} 1 & \sin\phi\tan\theta & \cos\phi\tan\theta \\ 0 & \cos\phi & -\sin\phi \\ 0 & \sin\phi\sec\theta & \cos\phi\sec\theta \end{bmatrix}\begin{bmatrix} p \\ q \\ r \end{bmatrix}$$

Dynamics

Translation

Using $V_g^b = [u\ v\ w]^T$, the velocity with respect to ground in the body frame

$$m\frac{d\vec{V}_g}{dt_i} = \underline{f}$$

we want in body frame $\Rightarrow = \dfrac{d\vec{V}_g}{dt_b} + \vec{\omega}_{b/i} \times \vec{V}_g$ } from earlier $\vec{\omega}_{b/i}$ is $[\dot\phi, \dot\theta, \dot\psi]^T$

Alternatively:

$$m\left(\frac{d\vec{V}_g^b}{dt_b} + \vec{\omega}_{b/i}^b \times \vec{V}_g^b\right) = \vec{f}^b$$ which is in the body frame, $V_g^b = [u, v, w]^T$, $W_{b/i}^b = [p, q, r]$

$$\frac{d\vec{V}_g}{dt_b} = \begin{bmatrix} \dot u \\ \dot v \\ \dot w \end{bmatrix} \Rightarrow \begin{bmatrix} \dot u \\ \dot v \\ \dot w \end{bmatrix} = \begin{bmatrix} rv - qw \\ pw - ru \\ qu - pv \end{bmatrix} + \frac{1}{m}\begin{bmatrix} f_x^b \\ f_y^b \\ f_z^b \end{bmatrix}$$

Rotational Motion

$$\frac{d\vec{L}^i}{dt_i} = \vec{\tau} = \frac{d\vec{L}}{dt_b} + \vec{\omega}_{b/i} \times \vec{L}$$

$\leftarrow$ angular momentum

Assuming moments taken about COM

Expressing in body frame:

$$\frac{d\vec{L}^b}{dt_b} + \vec{\omega}_{b/i}^b \times \vec{L}^b = \vec{\tau}$$

$$\underline{I} = \begin{bmatrix} I_{xx} & I_{xy} & I_{xz} \\ \ddots & I_{yy} & I_{yz} \\ \ddots & \ddots & I_{zz} \end{bmatrix}, \quad \frac{dI}{dt} = 0$$

$$\vec{L}^b = \underline{I}^b \vec{\omega}_{b/i} \Rightarrow \frac{d\vec{L}}{dt_b} = \underline{I}\frac{d\omega_{b/i}}{dt_b}$$

$$\Rightarrow \underline{I}\frac{d\vec{\omega}_{b/i}^b}{dt_b} + \vec{\omega}_{b/i}^b \times \left(\underline{I}\,\vec{\omega}_{b/i}^b\right) = \vec{\tau}^b$$

$$\dot{\vec{\omega}}_{b/i}^b = \begin{bmatrix} \dot p \\ \dot q \\ \dot r \end{bmatrix} \text{ and } \dot{\omega}_{b/i}^b = \underline{I}^{-1}\left[-\omega_{b/i}^b \times \left(\underline{I}\,\vec{\omega}_{b/i}^b\right) + \vec{\tau}^b\right]$$

Using Symmetry:

$$\underline{I} = \begin{bmatrix} I_x & 0 & -I_{xz} \\ 0 & I_y & 0 \\ -I_{xz} & 0 & I_z \end{bmatrix} \Rightarrow \underline{I}^{-1} = \begin{bmatrix} \frac{I_z}{\Gamma} & 0 & \frac{I_{xz}}{\Gamma} \\ 0 & \frac{1}{I_y} & 0 \\ \frac{I_{xz}}{\Gamma} & 0 & \frac{I_x}{\Gamma} \end{bmatrix} \quad \text{with } \Gamma = I_x I_z - I_{xz}^2$$

$$\Rightarrow \begin{bmatrix} \dot p \\ \dot q \\ \dot r \end{bmatrix} = \begin{bmatrix} \frac{I_z}{\Gamma} & 0 & \frac{I_{xz}}{\Gamma} \\ 0 & \frac{1}{I_y} & 0 \\ \frac{I_{xz}}{\Gamma} & 0 & \frac{I_x}{\Gamma} \end{bmatrix}\left(\begin{bmatrix} 0 & r & -q \\ -r & 0 & p \\ q & -p & 0 \end{bmatrix}\begin{bmatrix} I_x & 0 & -I_{xz} \\ 0 & I_y & 0 \\ -I_{xz} & 0 & I_z \end{bmatrix}\begin{bmatrix} p \\ q \\ r \end{bmatrix} + \begin{bmatrix} L \\ M \\ N \end{bmatrix}\right)$$

$$= \begin{bmatrix} \frac{I_z}{\Gamma} & 0 & \frac{I_{xz}}{\Gamma} \\ 0 & \frac{1}{I_y} & 0 \\ \frac{I_{xz}}{\Gamma} & 0 & \frac{I_x}{\Gamma} \end{bmatrix}\left(\begin{bmatrix} I_{xz}pq + (I_y - I_z)qr \\ I_{xz}(r^2 - p^2) + (I_z - I_x)pr \\ (I_x - I_y)pq - I_{xz}qr \end{bmatrix} + \begin{bmatrix} L \\ M \\ N \end{bmatrix}\right)$$

$$= \begin{bmatrix} \frac{I_{xz}(I_x - I_y + I_z)}{I_x I_z - I_{xz}^2}pq - \frac{I_z(I_z - I_y) + I_{xz}^2}{I_x I_y - I_{xz}^2}qr + \frac{I_z}{I_x I_z - I_{xz}^2}L + \frac{I_{xz}}{I_x I_z - I_{xz}^2}N \\ \frac{I_z - I_x}{I_y}qr - \frac{I_{xz}}{I_y}(p^2 - r^2) + \frac{M}{I_y} \\ \frac{(I_x - I_y)I_x + I_{xz}^2}{I_x I_z - I_{xz}^2}pq - \frac{I_{xz}(I_x - I_y + I_z)}{I_x I_z - I_{xz}^2}qr + \frac{I_{xz}}{I_x I_z - I_{xz}^2}L + \frac{I_x}{I_x I_z - I_{xz}^2}N \end{bmatrix}$$

$$= \begin{bmatrix} \frac{I_{xz}(I_x - I_y + I_z)}{I_x I_z - I_{xz}^2}pq - \frac{I_z(I_z - I_y) + I_{xz}^2}{I_x I_y - I_{xz}^2}qr \\ \frac{I_z - I_x}{I_y}qr - \frac{I_{xz}}{I_y}(p^2 - r^2) \\ \frac{(I_x - I_y)I_x + I_{xz}^2}{I_x I_z - I_{xz}^2}pq - \frac{I_{xz}(I_x - I_y + I_z)}{I_x I_z - I_{xz}^2}qr \end{bmatrix} + \begin{bmatrix} \frac{I_z}{I_x I_z - I_{xz}^2}L + \frac{I_{xz}}{I_x I_z - I_{xz}^2}N \\ \frac{M}{I_y} \\ \frac{I_{xz}}{I_x I_z - I_{xz}^2}L + \frac{I_x}{I_x I_z - I_{xz}^2}N \end{bmatrix}$$

1.) we can use $\begin{bmatrix} p_n \\ p_e \\ p_d \end{bmatrix} = \ldots$, $\begin{bmatrix} \dot u \\ \dot v \\ \dot w \end{bmatrix} = \ldots$, $\begin{bmatrix} \dot\phi \\ \dot\theta \\ \dot\psi \end{bmatrix} = \ldots$ and $\begin{bmatrix} \dot p \\ \dot q \\ \dot r \end{bmatrix} = \ldots$ to represent our aircraft

2.) Implement the different formulations using both hand calcs and the full vector equations

Re-expressing with Quaternions

Quaternion $\underline{e} \in \mathbb{R}^4$, $\underline{e} = e_0 + e_1\hat{i} + e_2\hat{j} + e_3\hat{k}$, $\hat{i}^2 = \hat{j}^2 = \hat{k}^2 = \hat{i}\hat{j}\hat{k} = -1 \}$ from this $\hat{i}\hat{j} = \hat{k} = -\hat{j}\hat{k}$

$\Rightarrow$ equivalently $\underline{e} = [e_0, \vec{\varepsilon}]^T$, $\underline{\varepsilon} = [e_1, e_2, e_3]^T$

If you rotate by θ about an axis $\vec{v}$, the quaternion is related via:

$$e_0 = \cos\left(\frac{\theta}{2}\right), \quad \underline{\varepsilon} = \vec{v}\sin\left(\frac{\theta}{2}\right) = \begin{bmatrix} e_1 \\ e_2 \\ e_3 \end{bmatrix}$$

A quaternion product is defined as:

$$\underline{a} \otimes \underline{b} = \begin{bmatrix} a_0 b_0 - \vec{\alpha}\cdot\vec{\beta} \\ a_0\vec{\beta} + b_0\vec{\alpha} + \underbrace{\vec{\alpha}\times\vec{\beta}}_{\text{makes it non commutative}} \end{bmatrix}, \text{ with } \underline{a} = [a_0, \vec{\alpha}]^T, \ \underline{b} = [b_0, \vec{\beta}]^T$$

To actually rotate a vector $\vec{r}$, it's expressed as $(0, \vec{r})$, and we sandwich between $\overbrace{\underline{e} \text{ and its conjugate } e^*}^{\text{(hence } \frac{\theta}{2} \text{ as its done twice)}}$ $(= [e_0, -\vec{\varepsilon}]^T)$

$$(0, \vec{r}^{\,v}) = \underline{e} \otimes (0, \vec{r}^{\,b}) \otimes \underline{e}^*, \quad (0, \vec{r}^{\,b}) = \underline{e}^* \otimes (0, \vec{r}^{\,v}) \otimes \underline{e}$$

$\Rightarrow$ $\underline{e} \otimes (\cdot) \otimes \underline{e}^*$ takes body components to vehicle components

$\underline{e}$ is the attitude of the body frame relative to vehicle (ie inertial without translation)

The rotation Matrix

expanding:

$$\underline{e} \otimes (0, \vec{r}) \otimes \underline{e}^* = \left[-\vec{\varepsilon}\cdot\vec{r}^{\,b}, \ e_0\vec{r}^{\,b} + \vec{\varepsilon}\times\vec{r}^{\,b}\right]^T \otimes [e_0, -\vec{\varepsilon}]^T = \vec{r}^{\,v}$$

$$= \left[-e_0(\vec{\varepsilon}\cdot\vec{r}^{\,b}) + (e_0\vec{r} + \vec{\varepsilon}\times\vec{r}^{\,b})\cdot\vec{\varepsilon}, \ (\vec{\varepsilon}\cdot\vec{r}^{\,b})\vec{\varepsilon} + e_0(e_0\vec{r} + \vec{\varepsilon}\times\vec{r}^{\,b}) + (e_0\vec{r} + \vec{\varepsilon}\times\vec{r}^{\,b})\times(-\vec{\varepsilon})\right]^T$$

and with $\vec{\varepsilon}\times(\vec{\varepsilon}\times\vec{r}) = \vec{\varepsilon}(\vec{\varepsilon}\cdot\vec{r}) - \vec{r}(\vec{\varepsilon}\cdot\vec{\varepsilon})$ (vector identity, very common when manipulating Navier-Stokes)

$$\Rightarrow \Big[\underbrace{-e_0(\vec{\varepsilon}\cdot\vec{r}) + e_0(\vec{r}\cdot\vec{\varepsilon})}_{\text{equal but opposite sign}} + \overset{0}{\underbrace{(\vec{\varepsilon}\times\vec{r})\cdot\vec{\varepsilon}}_{\varepsilon\times\vec{r} \text{ is } \perp \text{ to both } \vec{\varepsilon} \text{ and } \vec{r}, \ \therefore \text{ dot } = 0}}, \ (\vec{\varepsilon}\cdot\vec{r})\vec{\varepsilon} + e_0^2\vec{r} + \underbrace{e_0(\vec{\varepsilon}\times\vec{r}) - e_0(\vec{r}\times\vec{\varepsilon})}_{\substack{\vec{r}\times\vec{\varepsilon} = -\vec{\varepsilon}\times\vec{r} \\ = 2e_0(\vec{\varepsilon}\times\vec{r})}} - \underbrace{(\vec{\varepsilon}\times\vec{r})\times\vec{\varepsilon}}_{\substack{= +\vec{\varepsilon}\times(\vec{\varepsilon}\times\vec{r}) \\ = (\vec{\varepsilon}\cdot\vec{r})\vec{\varepsilon} - \underbrace{(\vec{\varepsilon}\cdot\vec{\varepsilon})}_{\varepsilon^T\varepsilon}\vec{r}}}\Big]^T$$

$$\Rightarrow \vec{r}^{\,v} = \underbrace{(e_0^2 - \vec{\varepsilon}^T\vec{\varepsilon})}_{(e_0^2 - \vec{\varepsilon}^T\vec{\varepsilon})\underline{I}}\vec{r}^{\,b} + \underbrace{2(\vec{\varepsilon}\cdot\vec{r}^{\,b})\vec{\varepsilon}}_{\substack{\text{regrouping + associativity} \\ \to 2(\varepsilon\varepsilon^T)}} + 2e_0(\vec{\varepsilon}\times\vec{r}^{\,b})$$

$$\vec{\varepsilon}\times\vec{r} = \begin{bmatrix} e_2r_3 - e_3r_2 \\ e_3r_1 - e_1r_3 \\ e_1r_2 - e_2r_1 \end{bmatrix} = \begin{bmatrix} 0 & -e_3 & e_2 \\ e_3 & 0 & -e_1 \\ -e_2 & e_1 & 0 \end{bmatrix}\vec{r}$$

$$\Rightarrow R_b^v(\underline{e}) = \begin{bmatrix} e_0^2 + e_1^2 - e_2^2 - e_3^2 & 2(e_1e_2 - e_0e_3) & 2(e_1e_3 + e_0e_2) \\ 2(e_1e_2 + e_0e_3) & e_0^2 - e_1^2 + e_2^2 - e_3^2 & 2(e_2e_3 - e_0e_1) \\ 2(e_1e_3 - e_0e_2) & 2(e_2e_3 + e_0e_1) & e_0^2 - e_1^2 - e_2^2 + e_3^2 \end{bmatrix} = R_v^b(e)^T$$

Quaternion Kinematics

Over interval dt, body will turn by angle $\|\vec{\omega}\|dt$ about axis $\frac{\vec{\omega}}{\|\vec{\omega}\|}$ (in body axes as $\vec{\omega} = [p, q, r]^T$)

With $\underline{e} = [\cos\frac{\theta}{2}, \vec{v}\sin(\frac{\theta}{2})]$

$\Rightarrow \delta\underline{e} \approx [1, \frac{1}{2}\vec{\omega}\,dt]$

rotating about current body axes will compose on the right

$\Rightarrow \underline{e}(t + dt) = \underline{e}(t) \otimes \delta\underline{e} = e(t) + e(t) \otimes (0, \frac{1}{2}\vec{\omega})dt$

$\Rightarrow \dot{\underline{e}} = \frac{1}{2}\underline{e} \otimes (0, \vec{\omega}) = \frac{1}{2}[0 - \vec{\varepsilon}\cdot\vec{\omega}, \ e_0\vec{\omega} + \vec{\varepsilon}\times\vec{\omega}]$

$$\Rightarrow \underset{\uparrow}{\begin{bmatrix} \dot{e}_0 \\ \dot{e}_1 \\ \dot{e}_2 \\ \dot{e}_3 \end{bmatrix}} = \frac{1}{2}\begin{bmatrix} 0 & -p & -q & -r \\ p & 0 & r & -q \\ q & -r & 0 & p \\ r & q & -p & 0 \end{bmatrix}\begin{bmatrix} e_0 \\ e_1 \\ e_2 \\ e_3 \end{bmatrix}$$

the magnitude will change, so we need to renormalise (or do some gradient descent lets see)